

PRACTICAL 1

MAHLATSE KGOLANE

Q1

```
--- INNER JOIN ORDERS WITH CUSTOMERS AND PRODUCT NAME
---E.O COLUMNS : ORDER_ID,CUSTOMERNAME,PRODUCT_NAME,QUANTITY
```

```
SELECT A.ORDERID,
       A. ORDERDATE,
       B. CUSTOMERNAME,
       C. PRODUCTNAME,
       A. QUANTITY
FROM PRACTICAL_2.LARGE.ORDERS AS A
INNER JOIN PRACTICAL_2.LARGE.CUSTOMERS AS B ON A.CUSTOMERID =B.CUSTOMERID
INNER JOIN PRACTICAL_2.LARGE.PRODUCTS AS C ON A.PRODUCTID = C.PRODUCTID;
```

	# ORDERID	🕒 ORDERDATE	📄 CUSTOMERNAME	📄 PRODUCTNAME	# QUANTITY
1	1	2023-06-10	Customer_1251	Product_2014	10
2	2	2023-12-07	Customer_1236	Product_2004	5
3	3	2024-10-26	Customer_1170	Product_2171	9
4	4	2023-02-17	Customer_1344	Product_2007	2
5	5	2024-11-06	Customer_1319	Product_2061	2
6	6	2024-11-23	Customer_1185	Product_2190	3

Q2

```
---INNER JOIN CUSTOMERS WHO PLACED ORDERS
---E.O COLUMNS : CUSTOMERID,CUSTOMERNAME,COUNTRY,ORDERID,ORDERDATE
```

```
SELECT C.CUSTOMERID,
       C. CUSTOMERNAME,
       C. COUNTRY,
       O.ORDERID,
       O.ORDERDATE
FROM PRACTICAL_2.LARGE.CUSTOMERS AS C
INNER JOIN PRACTICAL_2.LARGE.ORDERS AS O
ON C.CUSTOMERID = O .CUSTOMERID;
```

	# CUSTOMERID	📄 CUSTOMERNAME	📄 COUNTRY	# ORDERID	🕒 ORDERDATE
1	1251	Customer_1251	Germany	1	2023-06-10
2	1236	Customer_1236	Australia	2	2023-12-07
3	1170	Customer_1170	Germany	3	2024-10-26
4	1344	Customer_1344	Canada	4	2023-02-17
5	1319	Customer_1319	USA	5	2024-11-06
6	1185	Customer_1185	Australia	6	2024-11-23

Q3

```

---LEFT JOIN
--- E.O COLUMNS : CUSTOMERID,CUSTOMERNAME,COUNTRY,ORDERID,ORDERDATE,PRODUCTID,QUANTITY

SELECT C.CUSTOMERID,
       C.CUSTOMERNAME,
       C.COUNTRY,
       O.ORDERID,
       O.ORDERDATE,
       O.PRODUCTID,
       O.QUANTITY
FROM PRACTICAL_2.LARGE.CUSTOMERS AS C
LEFT JOIN PRACTICAL_2.LARGE.ORDERS AS O ON C.CUSTOMERID = O.CUSTOMERID;

```

	# CUSTOMERID	A CUSTOMERNAME	A COUNTRY	# ORDERID	🕒 ORDERDATE	# PRODUCTID	# QUANTITY
1	1251	Customer_1251	Germany	1	2023-06-10	2014	10
2	1236	Customer_1236	Australia	2	2023-12-07	2004	5
3	1170	Customer_1170	Germany	3	2024-10-26	2171	9
4	1344	Customer_1344	Canada	4	2023-02-17	2007	2
5	1319	Customer_1319	USA	5	2024-11-06	2061	2

Q4

```

---LEFT JOIN
---E.O COLUMNS : PRODUCT ID,PRODUCT NAME,TOTALORDERS

SELECT C.PRODUCTID,
       C.PRODUCTNAME,
       COUNT(O.ORDERID) AS TOTALORDERS
FROM PRACTICAL_2.LARGE.PRODUCTS AS C
LEFT JOIN PRACTICAL_2.LARGE.ORDERS AS O
ON C.PRODUCTNAME = C.PRODUCTNAME
GROUP BY C.PRODUCTID,C.PRODUCTNAME
ORDER BY C.PRODUCTID;

```

	# PRODUCTID	A PRODUCTNAME	# TOTALORDERS
1	2001	Product_2001	4000
2	2002	Product_2002	4000
3	2003	Product_2003	4000
4	2004	Product_2004	4000
5	2005	Product_2005	4000
6	2006	Product_2006	4000

Q5

```

SELECT O.ORDERID,
       O.ORDERDATE,
       C.PRODUCTID,
       C.PRODUCTNAME,
       C.PRICE,
       O.QUANTITY
FROM PRACTICAL_2.LARGE.ORDERS AS O
RIGHT JOIN PRACTICAL_2.LARGE.PRODUCTS AS C ON O.PRODUCTID = C.PRODUCTID
ORDER BY C.PRODUCTID,O.ORDERID;

```

	# ORDERID	🕒 ORDERDATE	# PRODUCTID	A PRODUCTNAME	# PRICE	# QUANTITY
1	298	2023-07-28	2001	Product_2001	833	9
2	427	2023-02-28	2001	Product_2001	833	6
3	680	2024-10-29	2001	Product_2001	833	5
4	715	2024-07-17	2001	Product_2001	833	3
5	765	2023-05-12	2001	Product_2001	833	6
6	1259	2023-08-29	2001	Product_2001	833	5

Q6

--E.O COLUMNS: CUSTOMERID,CUSTOMERNAME,COUNTRY,ORDERID,ORDERDATE,PRODUCTID,QUANTITY

```

SELECT C.CUSTOMERID,
       C.CUSTOMERNAME,
       C.COUNTRY,
       O.ORDERID,|
       O.ORDERDATE,
       O.PRODUCTID,
       O.QUANTITY
FROM PRACTICAL_2.LARGE.CUSTOMERS AS C
LEFT JOIN PRACTICAL_2.LARGE.ORDERS AS O
ON C.CUSTOMERID = O.CUSTOMERID
ORDER BY C.CUSTOMERID,O.ORDERDATE;

```

	# CUSTOMERID	A CUSTOMERNAME	A COUNTRY	# ORDERID	🕒 ORDERDATE	# PRODUCTID	# QUANTITY
1	1001	Customer_1001	India	2602	2023-04-04	2197	7
2	1001	Customer_1001	India	2802	2024-03-14	2005	6
3	1001	Customer_1001	India	3408	2024-04-15	2100	5
4	1001	Customer_1001	India	1855	2024-05-04	2185	7
5	1001	Customer_1001	India	1106	2024-05-15	2084	10

Q7

```

---FULL OUTER JOIN
---E.O COLUMNS : CUSTOMERID,CUSTOMERNAME,COUNTRY,ORDERID,RDERDATE,PRODUCTID,QUANTITY

```

```

SELECT C.CUSTOMERID,
       C.CUSTOMERNAME,
       C.COUNTRY,
       O.ORDERID,
       O.ORDERDATE,
       O.PRODUCTID,
       O.QUANTITY
FROM PRACTICAL_2.LARGE.CUSTOMERS AS C
FULL OUTER JOIN PRACTICAL_2.LARGE.ORDERS AS O ON C.CUSTOMERID = O.CUSTOMERID
ORDER BY COALESCE(C.CUSTOMERID,O.CUSTOMERID);

```

	# CUSTOMERID	A CUSTOMERNAME	A COUNTRY	# ORDERID	⌚ ORDERDATE	# PRODUCTID	# QUANTITY
1	1001	Customer_1001	India	2802	2024-03-14	2005	6
2	1001	Customer_1001	India	1855	2024-05-04	2185	7
3	1001	Customer_1001	India	3408	2024-04-15	2100	5
4	1001	Customer_1001	India	1106	2024-05-15	2084	10
5	1001	Customer_1001	India	127	2024-06-30	2190	5

Q8

```

SELECT C.PRODUCTID,
       C.PRODUCTNAME,
       C.PRICE,
       O.ORDERID,
       O.ORDERDATE,
       O.CUSTOMERID,
       O.QUANTITY
FROM PRACTICAL_2.LARGE.PRODUCTS AS C
FULL OUTER JOIN PRACTICAL_2.LARGE.ORDERS AS O
ON C.PRODUCTID = O.PRODUCTID
ORDER BY COALESCE(C.PRODUCTID,O.PRODUCTID);

```

	# PRODUCTID	A PRODUCTNAME	# PRICE	# ORDERID	⌚ ORDERDATE	# CUSTOMERID	# QUANTITY
1	2001	Product_2001	833	2180	2023-07-19	1249	3
2	2001	Product_2001	833	2723	2024-11-26	1368	5
3	2001	Product_2001	833	2544	2024-04-23	1186	6
4	2001	Product_2001	833	3555	2024-10-27	1358	10
5	2001	Product_2001	833	2485	2024-04-26	1284	1